# Net force first

**antocciphysics**

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Combine forces before using mass to predict acceleration.

**SUCCESS CRITERIA - I can...**

1. Find net force before dividing by mass.
2. State acceleration with units and direction.
3. Separate acceleration direction from motion direction.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Warm-up: bring back an earlier idea

A 4 kg object hangs still from a vertical string. Use g=9.8 N/kg. Name all forces on it with sources, sizes and directions. Does object-on-string belong on this diagram?

**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the model

**Reference:** ΣF = ma; a = ΣF/m.
Add signed forces on the same object before dividing by mass. N/kg = m/s².

**Worked reasoning:** A 2 kg cart has 9 N right and 3 N left. Net force=6 N right; a=6/2=3 m/s² right. Add forces before dividing by mass.

**Your turn:** Draw arrows for both forces, then a separate net-force arrow. Explain why 9/2 is not the acceleration.




2 kg cart: draw the net force separately.

**Marzano check - circle:** ① ② ③

## 03 Work from force arrows

A 5 kg object has 32 N right and 12 N left. It currently moves left at 6 m/s. Find a and v after 1 s under constant forces.




The dot represents the object. Mass: 5 kg; velocity left

**Marzano check - circle:** ① ② ③

## 04 Find an unknown force

A 3 kg cart accelerates left at 2 m/s². One force is 4 N right; the other horizontal force is unknown. Complete the diagram and find the unknown force magnitude.




Mass 3 kg; acceleration 2 m/s² left.

**Marzano check - circle:** ① ② ③

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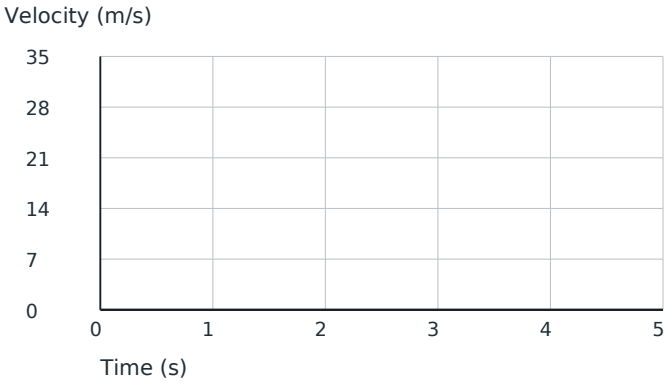
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Continue down each column.

## 05 From braking to force

A 1500 kg car brakes steadily from 30 m/s to rest in 5 s. State a positive direction, plot the velocity-time line, then find acceleration and net force.




**Marzano check - circle:** ① ② ③

## 06 Review: connect an earlier idea

Positions are 21, 14, 7 m at 0, 2, 4 s. Predict position at 9 s. Give the rate, an assumption and a future test. Separate what was observed from what can be inferred.

| Observation | Supported claim | Missing information |
|---|---|---|
| ___ | ___ | ___ |
| ___ | ___ | ___ |

**Marzano check - circle:** ① ② ③

## 07 Exit check

A 4 kg cart has 7 N right and 19 N left. Find a. Is it necessarily moving left? Name the information needed to determine whether it speeds up.



Add and label force arrows.

Draw the given forces; show net force separately.

**Marzano check - circle:** ① ② ③

## 08 STRETCH: test the model

A 3 kg cart accelerates left at 2 m/s². A rope pulls right with 10 N. Find the other horizontal force. Can the cart be moving right at this instant?



Add and label force arrows.

Label each interaction force on the cart.

**Marzano check - circle:** ① ② ③